SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1005	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:		**Reg. No.:**	
Date:		**Duration:**	60 minutes

Code of Conduct

I ___Mathivadhani K(192311424)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __Mathivadhani.K__

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an Agile Sprint Planning and User Story Workshop by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an Agile Sprint Planning and User Story Workshop by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format:
     As a <user>, I want <feature>, so that <benefit>.
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	**Mark**
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

AI-Powered Personalized Nutrition and Meal Planning Platform

Agile Sprint Planning and User Story Workshop

1. Project Vision and Stakeholders

1.1 Project Vision

For individuals who want healthy eating plans that suit their own preferences, allergies, lifestyle and fitness goals, and for nutritionists who guide them, the AI-Powered Personalized Nutrition and Meal Planning Platform is a web-based application that generates personalized meal plans, tracks daily food and water intake, and alerts users about nutritional deficiencies or excess consumption. Unlike generic diet applications, it considers individual constraints, local food availability and changing preferences, and connects users with nutrition professionals.

1.2 Project Objectives

- Create personalized nutritional profiles and record dietary preferences and restrictions.
- Generate customized meal plans that users can modify by food availability and preference.
- Track daily food consumption and monitor nutritional intake.
- Provide meal and hydration reminders and alerts for deficiencies or excess intake.
- Track progress toward nutritional goals and generate nutritional reports.
- Support nutritionist interaction and maintain secure user information.

1.3 Key Stakeholders

- **Users / Clients:** create profiles, enter preferences and allergies, follow meal plans, log food and water and view progress. They expect safe, personalized and easy-to-follow plans with timely alerts.
- **Nutritionists / Dietitians:** review client profiles, adjust meal plans, monitor progress and guide clients. They expect efficient tools to manage many clients at once.
- **Product Owner:** owns the product backlog, sets priorities and accepts completed stories, aiming for maximum value from every sprint.
- **Scrum Master:** facilitates the Scrum events, removes impediments and coaches the team to keep the sprint flowing smoothly.
- **Development Team:** designs, develops, tests and delivers the sprint increment, and expects clear stories, acceptance criteria and a realistic sprint scope.
- **System Administrator:** manages accounts, the food and nutrient database, security and backups, and expects a secure, reliable and maintainable system.
- **Platform Management:** defines the privacy policy, provides resources and reviews outcomes such as adoption and user satisfaction.

2. Epics and User Stories

2.1 Epics

The requirements of the platform are grouped into six epics:

- **E1 – Account and Profile Management:** secure registration, login, nutritional profile, preferences and allergies (US-01, US-02, US-03).

- **E2 – Personalized Meal Planning:** AI-generated meal plans, meal swapping by availability, scheduling and reminders (US-04, US-05, US-06).
- **E3 – Nutrition Tracking and Alerts:** food and water logging, nutrient monitoring, and deficiency and excess alerts (US-07, US-08).
- **E4 – Progress Monitoring and Reports:** progress visualization and nutritional reports (US-09, US-10).
- **E5 – Nutritionist Interaction:** plan review by nutritionists and user–nutritionist communication (US-11, US-12).
- **E6 – Administration and Security:** user management and the food and nutrient database (US-13).

2.2 User Stories

Format: As a <user>, I want <feature>, so that <benefit>.

- **US-01 (E1):** As a new user, I want to register and log in securely, so that my personal and health-related information is protected.
- **US-02 (E1):** As a user, I want to create my nutritional profile with age, height, weight, activity level and goal, so that the system can calculate my daily nutritional targets.
- **US-03 (E1):** As a user, I want to record my dietary preferences, allergies and disliked foods, so that recommended meals are safe and suitable for me.
- **US-04 (E2):** As a user, I want to receive an AI-generated personalized meal plan, so that I do not have to plan my meals manually.
- **US-05 (E2):** As a user, I want to swap meals based on food availability and my preferences, so that the plan is practical for my daily life.
- **US-06 (E2):** As a user, I want to schedule meals and receive meal and hydration reminders, so that I follow my plan on time.
- **US-07 (E3):** As a user, I want to log my daily food and water intake, so that I can monitor what I consume.
- **US-08 (E3):** As a user, I want to receive alerts for nutritional deficiencies or excessive consumption, so that I can correct my diet early.
- **US-09 (E4):** As a user, I want to view charts of my progress toward my nutrition goals, so that I stay motivated.
- **US-10 (E4):** As a user, I want to download nutritional reports, so that I can review or share my dietary history.
- **US-11 (E5):** As a nutritionist, I want to review and adjust my clients' meal plans, so that they receive professional guidance.
- **US-12 (E5):** As a user, I want to chat with my nutritionist, so that I can clear doubts quickly.
- **US-13 (E6):** As an administrator, I want to manage users and the food and nutrient database, so that the platform stays accurate and secure.

3. Acceptance Criteria

Acceptance criteria are written in the Given–When–Then format for the six user stories selected for Sprint 1.

US-01: Secure registration and login

- **Given** a new visitor is on the registration page, **when** they submit a valid name, email and password (minimum 8 characters with a number), **then** the account is created and a confirmation is shown within 3 seconds.
- **Given** a registered user is on the login page, **when** they enter correct credentials, **then** they are logged in and taken to the dashboard for their role (user, nutritionist or admin).
- **Given** a user has entered wrong credentials 5 times, **when** they try to log in again, **then** the account is locked for 15 minutes and a message is displayed.
- **Given** any account is stored in the database, **when** the password is saved, **then** only a salted hash is stored, never the plain password.

US-02: Create nutritional profile

- **Given** a logged-in user is on the profile page, **when** they enter age, gender, height, weight, activity level and goal and save, **then** the profile is stored and shown again when the page is reopened.
- **Given** a required field is empty or invalid (for example, height 0), **when** the user clicks Save, **then** an error message highlights the field and nothing is saved.
- **Given** a valid profile is saved, **when** the system processes it, **then** bMI and daily calorie, protein, carbohydrate and fat targets are displayed within 2 seconds.

US-03: Record preferences and allergies

- **Given** the preferences form is open, **when** the user selects diet type, allergies and disliked foods and saves, **then** the selections are saved to the user profile.
- **Given** a user has the allergy “peanuts” saved, **when** a meal plan is generated, **then** no meal containing peanuts is included (0 allergen violations).
- **Given** preferences already exist, **when** the user edits and saves them, **then** the changes are applied to the next generated plan.

US-04: AI-generated personalized meal plan

- **Given** a user has a complete profile and preferences, **when** they click Generate plan, **then** a 7-day plan with breakfast, lunch, snack and dinner is displayed within 5 seconds.
- **Given** a meal plan is generated, **when** the daily totals are calculated, **then** daily calories are within ±10% of the user’s target.
- **Given** the user’s diet type is vegetarian, **when** a plan is generated, **then** all meals in the plan are vegetarian.
- **Given** the profile is incomplete, **when** the user clicks Generate plan, **then** the system asks the user to complete the profile first.

US-05: Swap meals by availability and preference

- **Given** a meal is shown in the plan, **when** the user selects Swap, **then** at least 3 alternatives within ±15% calories are shown, excluding the user’s allergens.
- **Given** the user marks “paneer” as unavailable, **when** the plan is regenerated or a meal is swapped, **then** no meal containing paneer is suggested.
- **Given** a meal has been swapped, **when** the new meal is confirmed, **then** the daily nutrient totals are recalculated and shown immediately.

US-07: Log daily food and water intake

- **Given** the food log page is open, **when** the user searches "idli", selects it and enters the quantity, **then** the item is added with its calories, protein, carbohydrates and fat.
- **Given** items are logged during the day, **when** a new item is added, **then** daily totals and remaining targets update within 2 seconds.
- **Given** the user drinks water, **when** they click Add a glass, **then** the hydration counter increases by one, up to the daily goal.
- **Given** an item was logged by mistake, **when** the user deletes it, **then** the totals are recalculated without that item.

4. Product Backlog Prioritization

The backlog is ordered using three factors: business value (how much the story helps users reach nutrition goals), customer need (core requirement in the problem statement, tagged with MoSCoW priority) and dependency (stories that other stories rely on come first). For example, login and profile must exist before any meal plan can be generated, and a meal plan must exist before it can be swapped or tracked.

Rank	ID	User Story	Value	Depends On	Priority	Points
1	US-01	Secure registration and login	High	–	Must	5
2	US-02	Create nutritional profile	High	US-01	Must	3
3	US-03	Record preferences and allergies	High	US-01	Must	3
4	US-04	AI-generated personalized meal plan	High	US-02, US-03	Must	8
5	US-05	Swap meals by availability and preference	High	US-04	Must	5
6	US-07	Log daily food and water intake	High	US-01, US-02	Must	5
7	US-08	Alerts for deficiency or excess intake	High	US-02, US-07	Must	8
8	US-06	Meal scheduling and hydration reminders	Medium	US-04	Should	3
9	US-09	Progress charts toward goals	Medium	US-07	Should	5
10	US-11	Nutritionist reviews and adjusts plans	Medium	US-04	Should	8
11	US-10	Download nutritional reports	Medium	US-07, US-09	Should	5
12	US-13	Admin manages users and food database	Medium	US-01	Should	8
13	US-12	Chat with nutritionist	Low	US-11	Could	5

Total backlog: 13 stories, 71 story points.

5. Sprint Backlog

5.1 Sprint Details

- **Sprint:** Sprint 1 of two weeks (10 working days), from 26 October 2026 to 6 November 2026.
- **Team:** one Product Owner, one Scrum Master and four Developers / Testers.
- **Capacity:** 4 members × 10 days × 5 productive hours = 200 hours.
- **Stories selected:** US-01, US-02, US-03, US-04, US-05 and US-07, all high-priority "Must" stories that form the base of the system.

- **Commitment:** 29 story points and 167 task hours, about 84% of capacity, which leaves a buffer for bugs, reviews and meetings.

5.2 Task Breakdown

Task	Story	Task Description	Hours	Owner
T1	US-01	Design login and registration UI	6	UI / Dev 1
T2	US-01	Build authentication API with password hashing and tokens	10	Dev 1
T3	US-01	Implement role-based access (user, nutritionist, admin)	8	Dev 1
T4	US-01	Unit tests for authentication and access control	6	Tester
T5	US-02	Design profile form and database schema	6	Dev 2
T6	US-02	Implement profile API with validation	8	Dev 2
T7	US-02	Calculate BMI and daily calorie and macronutrient targets	6	Dev 2
T8	US-02	Test profile saving and calculations	4	Tester
T9	US-03	Build preferences and allergy selection UI	5	Dev 3
T10	US-03	Store preferences and allergies in the database	5	Dev 2
T11	US-03	Validation and tests for preferences	3	Tester
T12	US-04	Prepare food and nutrient dataset including local foods	10	Dev 4
T13	US-04	Develop meal recommendation logic (rules and scoring)	16	Dev 3
T14	US-04	Apply allergy and preference filters	8	Dev 3
T15	US-04	Build meal plan UI (daily and weekly views)	10	Dev 4
T16	US-04	Test plan accuracy against calorie targets and allergies	8	Tester
T17	US-05	Develop swap and alternatives API	8	Dev 4
T18	US-05	Build availability toggles UI	6	UI / Dev 1
T19	US-05	Recalculate daily totals after a swap	4	Dev 4
T20	US-05	Test swapping and availability rules	4	Tester
T21	US-07	Build food log UI with search	8	Dev 1
T22	US-07	Develop food and water logging API	8	Dev 2
T23	US-07	Calculate daily nutrient totals against targets	6	Dev 2
T24	US-07	Test logging, deletion and totals	4	Tester
		Total sprint effort	167	

Task hours per story: US-01 = 30, US-02 = 24, US-03 = 13, US-04 = 52, US-05 = 22, US-07 = 26.

6. Story Point Estimation

Estimation was done using Planning Poker with the modified Fibonacci sequence (1, 2, 3, 5, 8, 13). Each team member privately selected a card based on effort, complexity and risk, and the team discussed differences until it agreed on one value. US-03 (a simple form with storage) was used as the reference story with 3 points; every other story was compared with it.

ID	User Story	Points	Justification
US-01	Secure registration and login	5	Well-known work but security-sensitive: hashing, token handling, role-based access and lockout need careful testing.
US-02	Create nutritional profile	3	Simple form and validation with a standard BMI and calorie formula. Similar to the reference story.

ID	User Story	Points	Justification
US-03	Record preferences and allergies	3	Reference story. Form and storage with multi-select inputs and little business logic.
US-04	AI-generated meal plan	8	Most complex story: food dataset, recommendation logic, allergy filters and plan UI. High uncertainty and dependency on US-02 and US-03.
US-05	Swap meals by availability	5	Needs alternative search with calorie and allergen matching, a toggle UI and recalculation of totals.
US-07	Log daily food and water intake	5	Search, logging, deletion, hydration counter and total calculations across several screens.

Total committed: 5 + 3 + 3 + 8 + 5 + 5 = 29 story points.
A story larger than 8 points was not selected, because large stories carry more uncertainty and are split into smaller ones. The commitment of 29 points matches the expected velocity of a new four-member team and the 200-hour capacity.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

Deliver secure registration and profile setup, and a personalized, adjustable meal plan with daily food and water logging, so that a user can sign up, enter their details and start following a safe meal plan.

7.2 Expected Deliverables at the End of Sprint 1

- Working registration, login and role-based access (US-01).
- Nutritional profile with calculated BMI and daily targets (US-02).
- Preferences and allergy management (US-03).
- AI-generated 7-day personalized meal plan that respects allergies and diet type (US-04).
- Meal swapping and food availability settings (US-05).
- Daily food and water logging with nutrient totals (US-07).
- Tested, demonstrable increment with updated product backlog and sprint review notes.

7.3 Definition of Done

- Code is reviewed and merged.
- Unit and integration tests pass.
- All acceptance criteria of the story are met.
- The feature is demonstrated to the Product Owner and accepted.

7.4 Scrum Events in the Sprint

- Sprint Planning on day 1 to agree the goal and scope.
- Daily Scrum of 15 minutes to share progress and blockers.
- Sprint Review on the last day to demonstrate the increment.
- Sprint Retrospective after the review to improve the way of working.

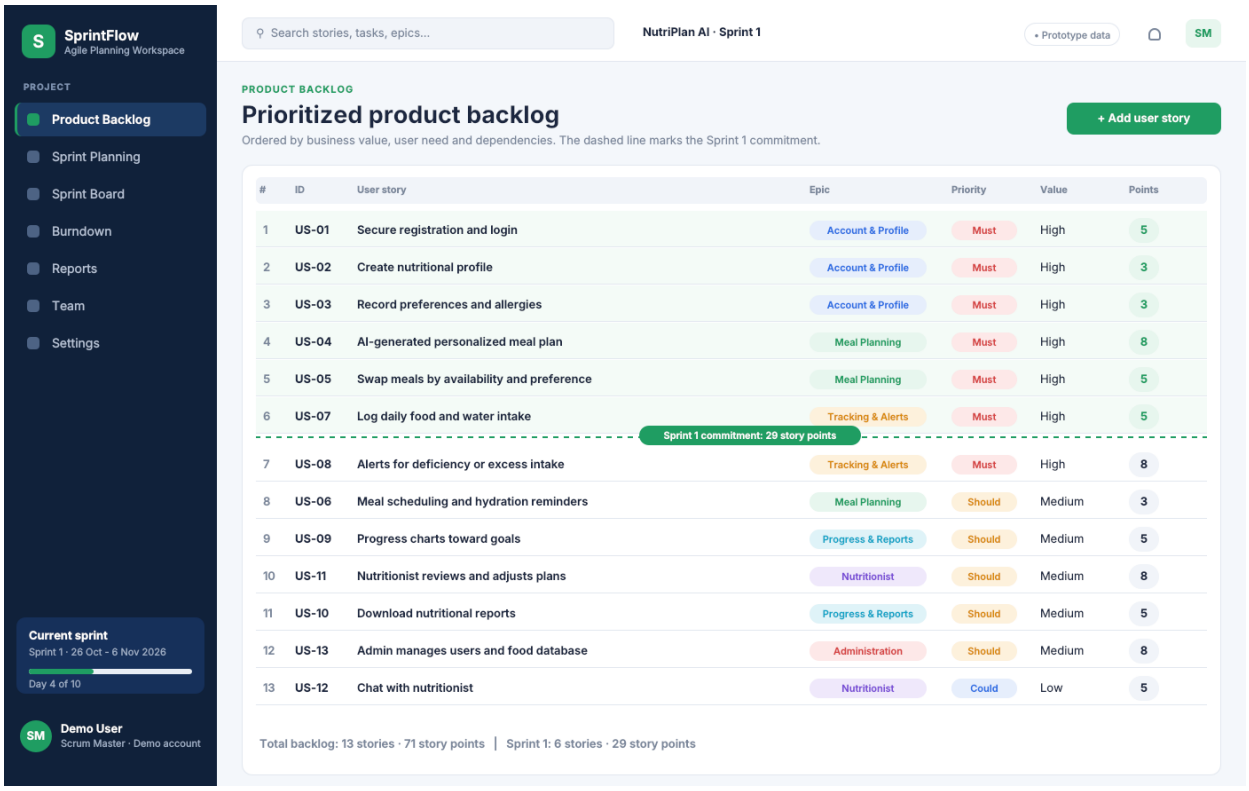
Conclusion

The sprint planning workshop turns the nutrition platform problem into a clear vision, six epics and thirteen user stories with measurable acceptance criteria. A prioritized backlog guided the selection of six high-priority stories (29 story points) for Sprint 1, broken into 24 tasks within the team's

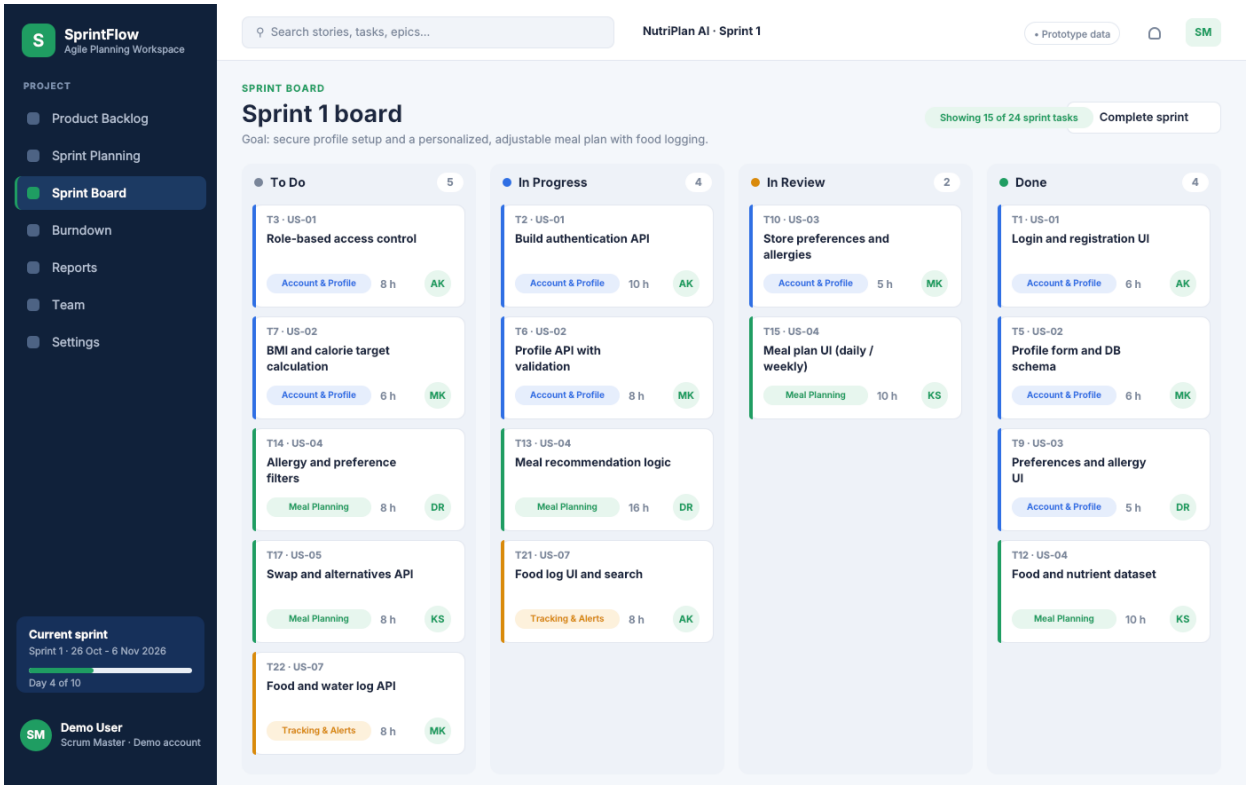
capacity. The sprint goal and deliverables give the team a realistic target and give stakeholders a working, testable increment at the end of the sprint.

SPRINT PLANNING OUTPUT SCREENSHOTS

PRODUCT BACKLOG (PRIORITIZED)



SPRINT 1 BOARD (TASK BREAKDOWN)



SPRINT PLAN: STORY POINTS, CAPACITY AND BURNDOWN

